

PERSONAL INFORMATION

Al Shaima Al Rawahi



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Highly analytical and motivated MSc graduate in Astrophysics with a robust foundation in data science, software development, and complex system analysis. Proven ability to design and implement data processing pipelines, apply machine learning for signal classification, and perform statistical modelling. Seeking a role to apply my strong Python programming, problem-solving, and data analysis skills.

EDUCATION

2023 – 2025	Master of Science in Astrophysics and Cosmology	105/110
	University of Padova – Padova, Italy	
	- Computational Focus: Gained proficiency in designing Python-based data pipelines, a skill directly transferable to creating structured lab exercises and computational physics projects for students. - Thesis: <i>Multiple Stellar Populations in the Globular Cluster NGC 6440</i> – Analysis of HST WFC3/UVIS imaging to investigate cluster kinematics and internal dynamics.	
2019 – 2022	Bachelor of Science in Physics with Astronomy	Second class division one
	University of Liverpool – Liverpool, United Kingdom	
	Thesis: <i>"Measuring the Shapes of Globular Clusters with Gaia"</i> – Independently processed Gaia DR3 data, wrote custom Python scripts, and produced a scientific report and presentation, demonstrating the ability to guide students through independent research projects.	

TEACHING & OUTREACH
EXPERIENCE

Invited Lecturer & Workshop Facilitator | March 2026
Oman Society of Astronomy and Space

- Designed and delivered an online workshop titled *"How We Discover What Cannot Be Seen: A Journey to Understand Black Holes and Star Clusters"* for society members and the public.
- Translated complex astrophysical concepts (light curves, stellar dynamics, black hole detection) into accessible, engaging content for a non-specialist audience.
- Received official certificate of appreciation from the Oman Astronomical Society for valuable contributions to public science education.

Graduate Research Mentor (Informal) | 2023 - 2025
University of Padova

- Collaboratively worked with thesis advisors and fellow graduate students, effectively training peers in data-reduction techniques, Python-based analysis (PSF photometry, astrometric calibration), and the use of scientific software (Fortran tools, Astropy).
- Developed clear documentation and step-by-step workflows for data pipelines, mirroring the creation of robust lab manuals and instructional materials.

RESEARCH & LABORATORY EXPERIENCE

Note: The following experiences demonstrate my hands-on proficiency with the scientific method, data analysis, and technical problem-solving.

Hubble Space Telescope Data Analysis | *Thesis Project, 2025*
University of Padova

- Conducted an independent research project reducing and aligning HST WFC3/UVIS images to study stellar populations and kinematics in NGC 6440.
- Developed a comprehensive Python-based data pipeline for distortion correction, PSF photometry, and astrometric calibration, demonstrating the ability to manage complex, multi-step scientific workflows.
- Produced high-precision stellar catalogues and performed proper-motion analysis to search for black-hole candidates, an end-to-end research process that can be adapted to design advanced undergraduate lab exercises in data analysis and the scientific method.

Computational Astrophysics & Exoplanet Transit Analysis | 2024
University of Padova

- Applied supervised machine-learning (SVM, Neural Networks) and Bayesian MCMC fitting to detect exoplanetary transits and model planetary parameters.
- Gained expertise in designing, validating, and interpreting computational models, skills directly applicable to developing project-based curricula for courses in computational physics and data science.
- Maintained clear, reproducible code and documentation for all data-preprocessing and analysis steps (Python, scikit-learn, emcee, batman).

High-Energy Astrophysics Laboratory | 2023
University of Padova

- Performed UV and X-ray data reduction of the interacting galaxy system M51 using professional-grade pipelines (Swift UVOT/XRT, HEASoft, XSPEC).
- Modelled X-ray spectra to derive physical properties like photon index and luminosity, an experience that provides a foundation for teaching multi-wavelength astronomy and spectroscopic techniques.
- Collaborated effectively with lab partners to co-add images, produce light curves, and interpret complex datasets, simulating a team-based research environment.

Globular Cluster Morphology with Gaia | *Undergraduate Research, 2022*
University of Liverpool

- Independently processed Gaia DR3 data, writing custom Python scripts for crossmatching, statistical analysis, and data visualization to quantify cluster ellipticities.
- Culminated the project in a scientific report and presentation.

PERSONAL SKILLS

Mother tongue Arabic

Other languages

English Proficient

Technical skills

Programming & Data Analysis	Modelling & Optimization	Data Processing	Software Tools
· Python (NumPy, Pandas, Matplotlib, Scikit-learn).	· Machine Learning (SVM, NN) · Bayesian Inference & MCMC · Model Validation & Error Analysis	· Sensor Data Calibration & Reduction	· Git / GitLab Version Control · Linux / Bash Scripting

Key Competencies

- Analytical Rigor: Proven ability to handle raw data and transform it into actionable scientific insights.
- Adaptability: Successfully transitioned between UK and Italian academic systems; capable of pivoting from astrophysics to environmental applications.
- Scientific Integrity: Strict adherence to documentation standards and research ethics.

Additional supporting documents are available upon request.